

Clinical Reasoning Analysis

A. Where Clinical Reasoning is Already Present

Within the current Clinical Teaching Project, an element of clinical reasoning is already explicitly integrated into Step 3 (Comprehension) of Peyton's Four-Step Approach (Krautter et al., 2011) during the paracentesis simulation. In this specific step, the third-year medical student does not physically perform the procedure. Instead, the learner verbally guides the faculty instructor through each of the technical steps of a diagnostic paracentesis. The instructor conducts the physical movements based on the student's verbal directions. The student needs to be precise in their instructions to include safe landmarking, enforcing sterile techniques, and use of the Z-track technique for needle insertion for example.

The specific teaching moment above primarily reflects the cognitive aspect of clinical reasoning where learners have to gather, interpret, and synthesize based on procedural script recollection. This is particularly relevant to both the diagnostic paracentesis component, but also to the ultrasound activity as well. Shifting the role of procedure execution to the faculty, it forces the learner to adapt and 'flips the script' allowing them to play the role of instructor. They must retrieve the sequential steps of a paracentesis from memory, anticipate the consequences of their verbal instructions, and engage in metacognitive monitoring to ensure that no steps from the checklist are omitted.

B. Where Clinical Reasoning Could be Strengthened

One area to develop clinical reasoning further is in learning objective 3 where a student is asked to differentiate between various clinical examples of ascitic fluid results. The activity was initially planned as three written clinical cases for students to review and calculate the serum-

ascites to albumin gradient and interpret the provided fluid studies. The gap in this design is both cognitive and narrative, as students are provided with the majority of the information and do not need to request or collect it on their own. From a cognitive perspective, this layout bypasses self-directed information gathering, initial hypothesis generation and independent problem representation (Elvén et al., 2023). From a narrative perspective, presenting isolated laboratory metrics disconnects the diagnostic data from the patient's clinical history.

This dual absence is notable as it would lead students to engage in superficial pattern matching rather than engage in true clinical reasoning. According to dual-process theory, clinical reasoning relies on an interplay between fast, automatic, non-analytical processes (System 1) and slow, rule-governed, analytical processes (System 2) (Pelaccia et al., 2011). By presenting static lab metrics without a narrative context, the activity does not allow learning to engage in System 2-type processes.

To deepen clinical reasoning, this session can be changed to integrate a sandboxed electronic medical record (EMR) with prepopulated sample patient charts. This mechanism already exists for small group sessions in the 2nd year of the medical school curriculum and would lend itself well to adaptation.

Under this new approach, initial admission notes would exist in the EMR along with basic initial laboratory results. Students must review the patient's clinical history, formulate their own initial differential diagnosis, and explicitly order the specific serum and ascitic fluid studies required. Once they have conducted the SAAG and analyzed their requested studies, they must then propose an assessment and plan which can be presented to the faculty towards the end of the simulation session.

This change introduces dual-process reasoning (Pelaccia et al., 2011) by combining non-analytical pattern recognition with analytical data processing. This therefore changes the student from being a passive learner to an active clinician within a safe, simulated clinical environment. Both System 1 and System 2 reasoning are integrated through this approach. Ultimately, integrating EMR-driven data gathering addresses a recognized deficit in health professions training (Elvén et al., 2023) by utilizing teaching and assessment methods that replicate authentic clinical environments to bridge simulated activities into clinical competence.

References

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